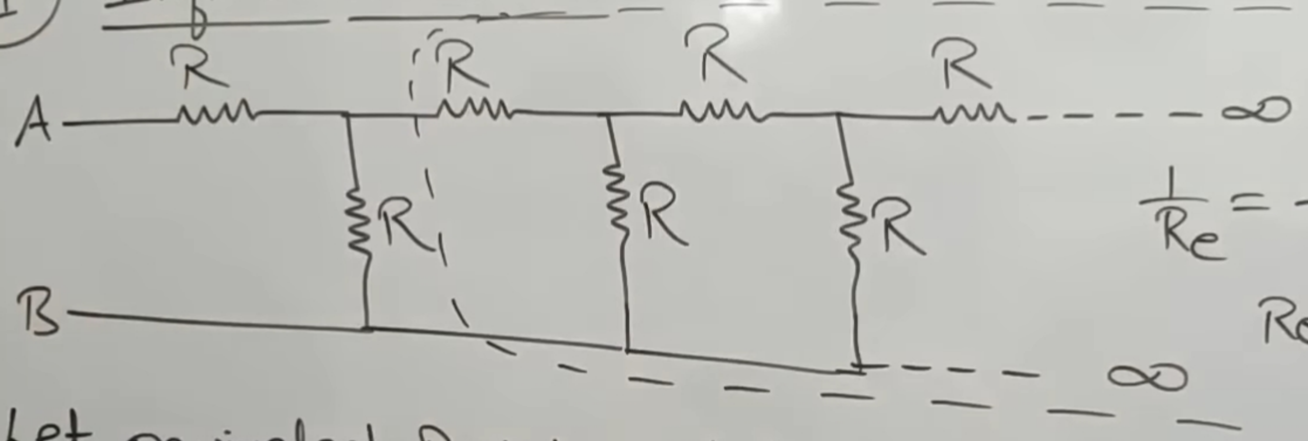
Combination of Resistors.

Find equivalen Resistance between A & B.

(II) Infinite Network

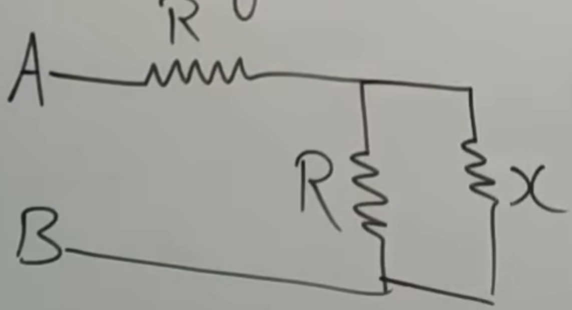


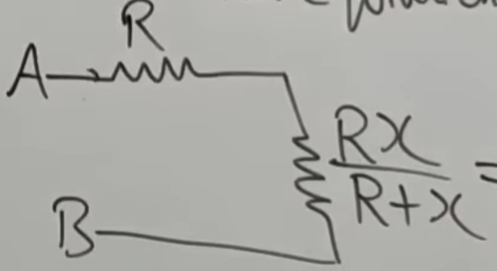
$$\frac{1}{R_e} = \frac{1}{R} + \frac{1}{x} = \frac{x+R}{Rx}$$

$$R_e = \frac{Rx}{R+x}$$

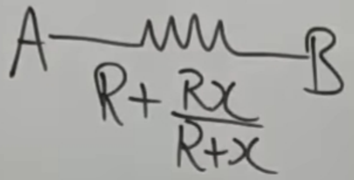
Let equivalent Resistance of this combination between A-B be 'x'.

ek Repeating unit ko chor ke baki ka equivalent Resistance $\rightarrow x$



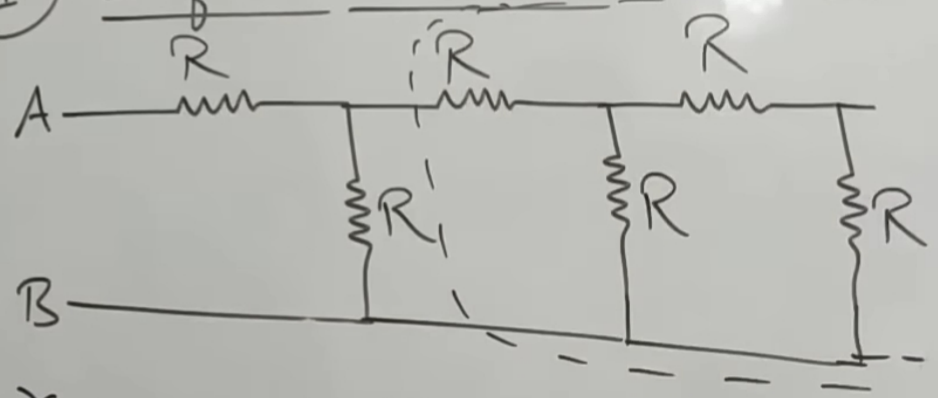


$\Rightarrow$



Combination of Resistors.

II) Infinite Network



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{+R \pm \sqrt{R^2 + 4R^2}}{2} = \frac{R \pm R\sqrt{5}}{2}$$

$$R + \frac{Rx}{R+x} = x$$

$$\frac{R(R+x) + Rx}{R+x} = x$$

$$R(R+x) + Rx = x(R+x)$$

$$R^2 + \cancel{Rx} + Rx = \cancel{Rx} + x^2$$

$$x^2 - Rx - R^2 = 0$$

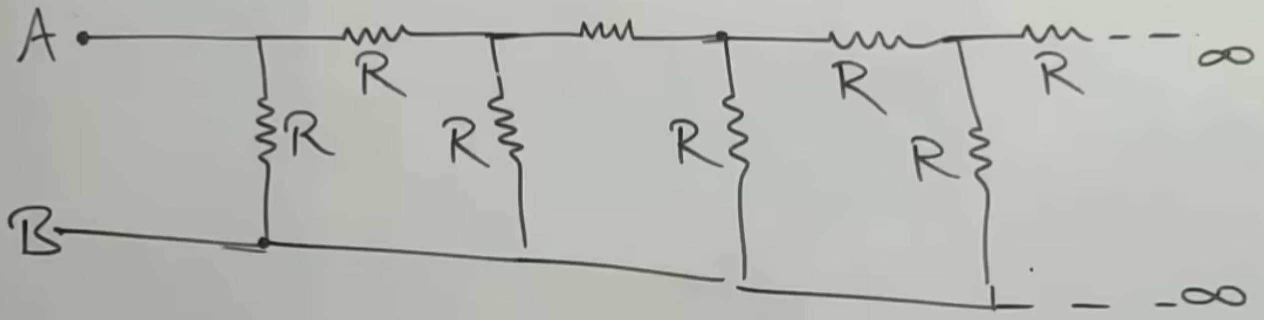
$$x = \frac{R - R\sqrt{5}}{2} \times \rightarrow -ve \times$$

$$x = \frac{R + R\sqrt{5}}{2} = \frac{R(\sqrt{5}+1)}{2}$$

Combination of Resistors.

II Infinite Network.

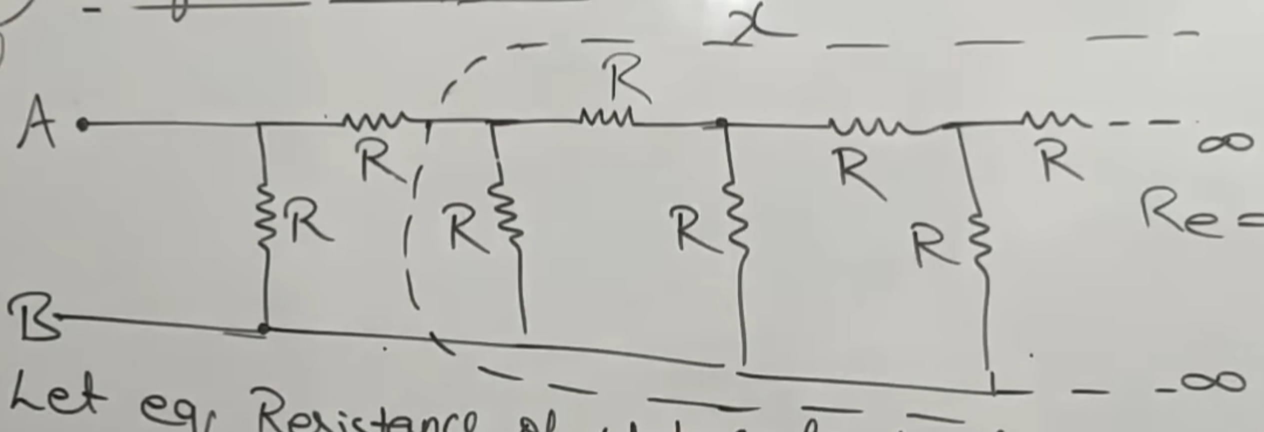
(2)



Combination of Resistors.

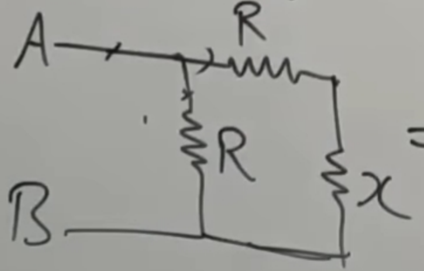
(II) Infinite Network.

(2)

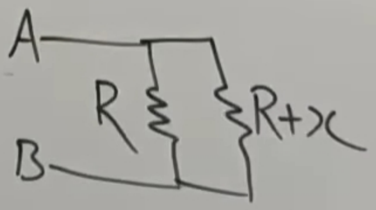


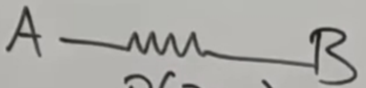
$$R_e = \frac{R_1 R_2}{R_1 + R_2}$$

Let eq Resistance of whole combination between A-B be 'x'
ek repeating unit chor K baki ka eq Resistance ⟶ x



$\Rightarrow$

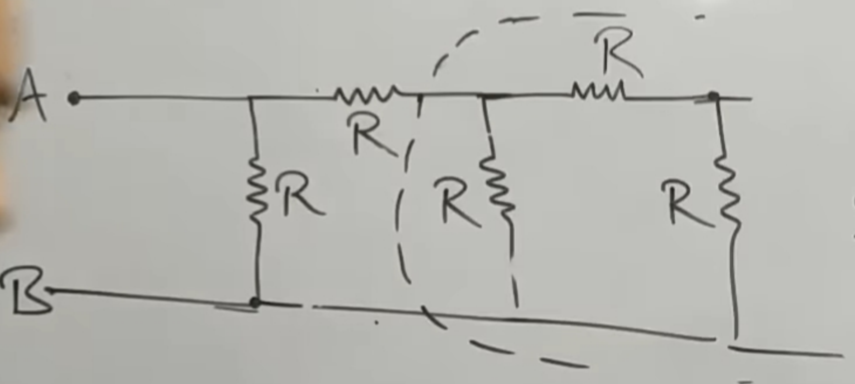




$$\frac{R(R+x)}{R+R+x}$$

Combination of Resistors.

Infinite Network.



$$\frac{R(R+x)}{R+(R+x)} = x$$

$$R^2 + Rx = x(2R+x)$$

$$R^2 + Rx = 2Rx + x^2$$

$$x^2 + Rx - R^2 = 0$$

$$x = \frac{-R \pm \sqrt{R^2+4R^2}}{2}$$

$$x = \frac{-R \pm \sqrt{R^2+4R^2}}{2}$$

$$= \frac{-R \pm R\sqrt{5}}{2}$$

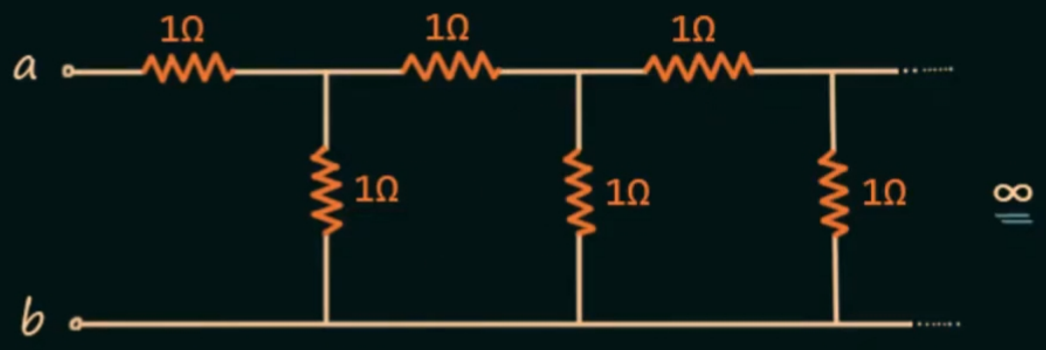
$$x = \frac{-R - R\sqrt{5}}{2} \quad \times\ -ve$$

$$x = \frac{-R + R\sqrt{5}}{2} = \frac{R(\sqrt{5}-1)}{2}$$

Equivalent Resistance (Solved Problem 3)

Question: Find the equivalent resistance between the terminals a–b.

Infinite Ladder N/w



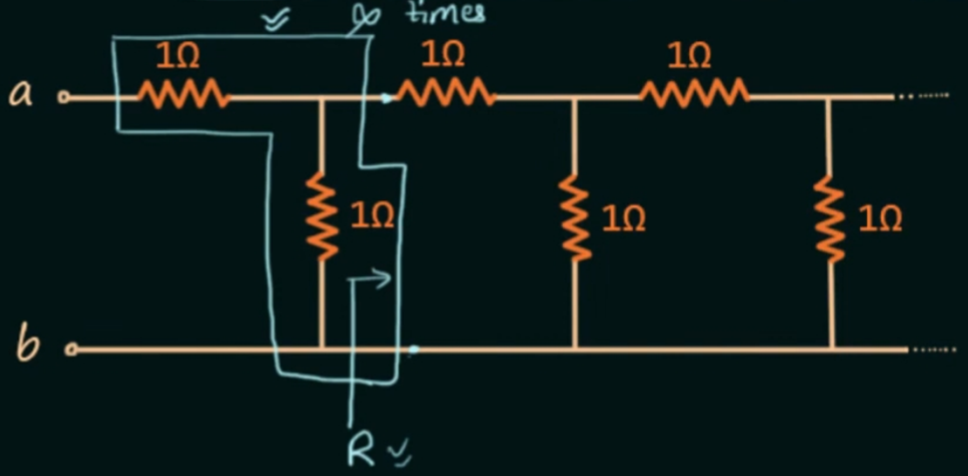
Solution:

Equivalent Resistance (Solved Problem 3)

Question: Find the equivalent resistance between the terminals a–b.

Infinite Ladder N/w

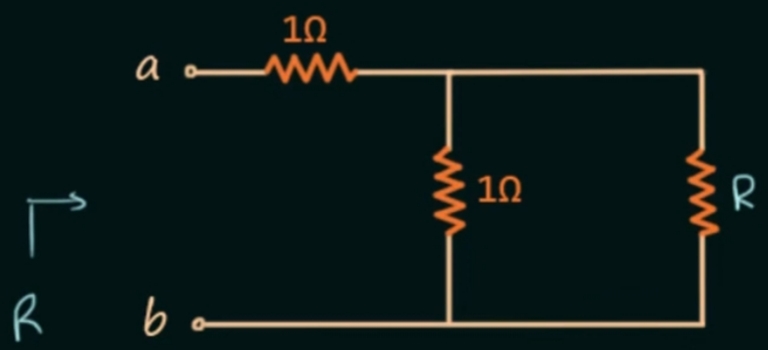
$R = ?$



$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Solution:



$$R = (R \parallel 1) + 1$$

$$R = \frac{R}{R+1} + 1$$

$$R(R+1) = R + (R+1)$$

$$R^2 + R = 2R + 1$$

$$R^2 - R - 1 = 0$$

$a = 1$, $b = -1$ & $c = -1$

$$R = \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \cdot 1 \cdot (-1)}}{2 \cdot 1}$$

$$= \frac{1 \pm \sqrt{1+4}}{2}$$

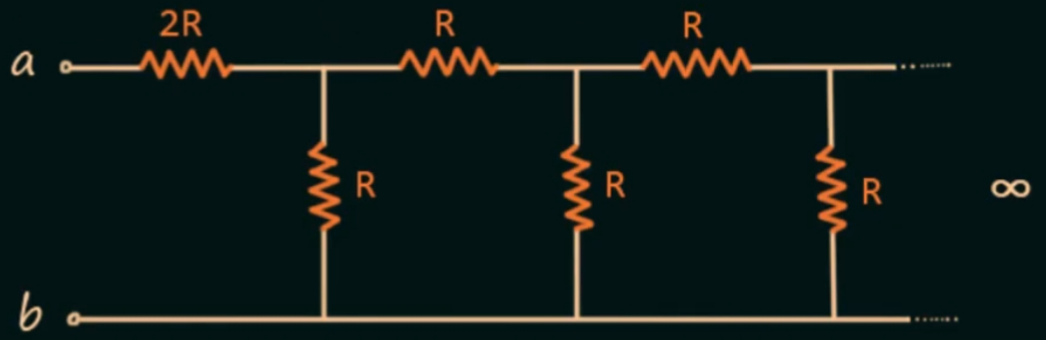
$$R = \frac{1 \pm \sqrt{5}}{2}$$

$$R = \frac{1+\sqrt{5}}{2}, \quad \boxed{\frac{1-\sqrt{5}}{2}}$$

$-ve$ >1

$$R = \frac{1+\sqrt{5}}{2}\ \Omega$$

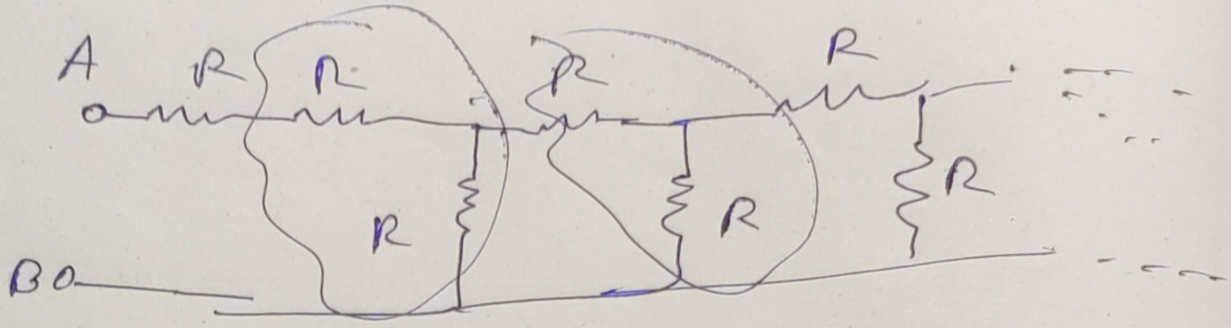
Homework: The equivalent resistance in the infinite ladder network shown in figure, is R_e

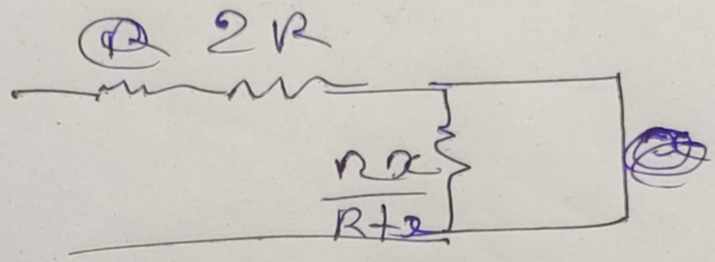


[GATE - 2014]

The value of R_e/R is _______ .

$\frac{\sqrt{5}+1}{2}$





$$x = 2R + \frac{Rx}{R+x}$$

$$x = \frac{2R(R+x) + Rx}{R+x}$$

$$\therefore Rx + x^2 = 2R^2 + 3Rx \cancel{+Rx}$$

$$\therefore x^2 - 2Rx - 2R^2 = 0$$

$$a = 1,\ b = -2R,\ c = -2R^2$$

$$4R^2 + 8R^2$$

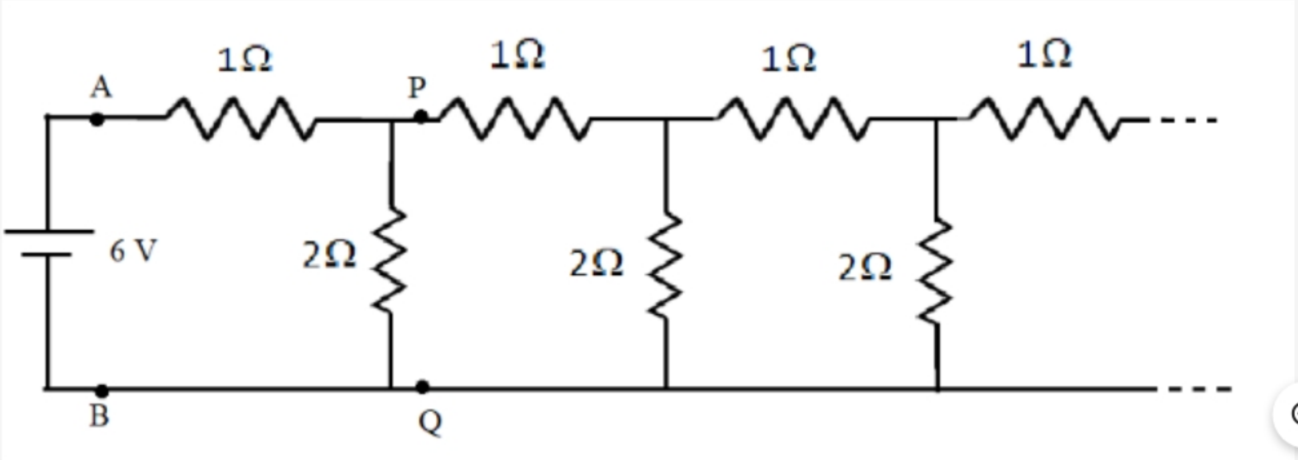
$$x = \frac{2R \pm \sqrt{12R^2}}{2}$$

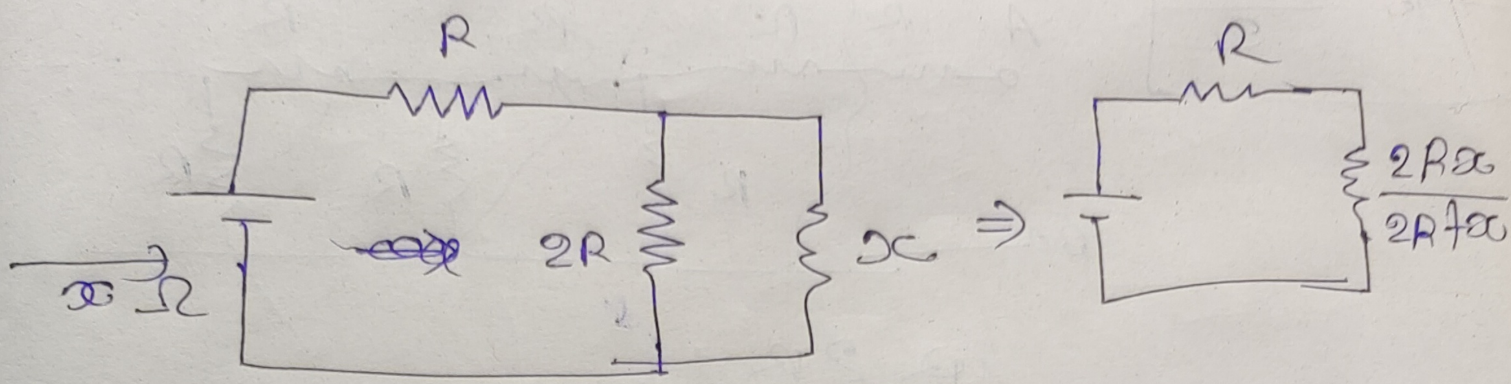
$$= \frac{2R \pm 2R\sqrt{3}}{2} = \frac{R + R\sqrt{3}}{2}$$

$$= \frac{R(1+\sqrt{3})}{2}$$

$$\therefore \boxed{x = \frac{R(1+\sqrt{3})}{2}}$$

An infinite ladder network of resistances is constructed with $1\,\Omega$ and $2\,\Omega$ resistances as shown in the figure. The 6 V battery between A and B has negligible internal resistance. The equivalent resistance between A and B is?





$$\therefore x = R + \frac{2Rx}{2R+x}$$

$$\therefore x(2R+x) = R(2R+x) + 2Rx$$

$$\therefore 2Rx + x^2 = 2R^2 + Rx + \cancel{2Rx}$$

$$\therefore x^2 - Rx - 2R^2 = 0$$

$$a = 1, \; b = -R, \; c = -2R^2$$

$$x = \frac{R \pm \sqrt{R^2 - 4(1)(-2R^2)}}{2}$$

$$= \frac{R \pm \sqrt{9R^2}}{2}$$

$$= \frac{4R}{2}$$

$$x = 2R$$

$$\therefore \boxed{x = 2(1) = 2\ \Omega}$$